Marvin Gülhan

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Education

University of Bonn — Bonn, Germany
M.Sc. Computer Science · Focus: AI & Algorithms · Current grade: 1.1 (1.0 = best) — *Oct. 2025 – Present*

Universidad Politécnica de Madrid — Madrid, Spain
M.Sc. Computer Science · Upcoming Exchange Semester — *Feb. 2027 – Jun. 2027*

University of Bonn — Bonn, Germany
B.Sc. Computer Science · Focus: Algorithms · Graduated with honors · Grade: 1.2 (1.0 = best) — *Oct. 2022 – Sep. 2025*

- Thesis (Grade: 1.0): adapted Hybrid Genetic Search (PyVRP) to the Dynamic VRPTW. Outperformed three state-of-the-art ACS baselines on the majority of dynamic Solomon instances

Rhein-Sieg-Gymnasium — Sankt Augustin, Germany
Abitur (Grade: 1.0) · Semester Abroad at GISSV (CA), USA — *Aug. 2014 – June 2022*

Experience

AI & Operations Research Intern — Bonn, Germany
DHL Group · Deutsche Post AG — *Oct. 2026 – Dec. 2026*

- Optimization of delivery-district partitioning and development of an AI agent-based natural-language query system over internal operational data

Research Assistant — Bonn, Germany
Fraunhofer Institute FKIE — *Nov. 2023 – Sep. 2026*

- Developed online combinatorial optimization methods for dynamic routing and resource allocation, including the Airlift Planning Problem
- Contracted Germany-wide OpenStreetMap road graphs of 10M+ edges into solver-ready instances for a national disaster-logistics pipeline
- Implemented heuristic baselines for GNN-based Reinforcement Learning policies on dynamic planning problems

Aerodynamics Team Member – Formula Student — Sankt Augustin, Germany
Bonn-Rhein-Sieg University of Applied Sciences (HBRS) — *Sep. 2023 – Jun. 2025*

- Designed and optimized the rear-wing aerodynamic package for the team's 2024 race car through iterative CFD simulations using OpenFOAM

Publications & Projects

Reinforcement Learning Amplifies Emergent Misalignment from Harmless Rewards — *May 2026*

- Submitted to EMNLP 2026
- Implemented and evaluated mitigation techniques against RL-induced emergent misalignment

Multi-Agent Reinforcement Learning for Complex and Dynamic Graph-Based Planning Problems — *May 2026*

- Presented at ICMCIS 2026
- Co-developed and evaluated a Graph Reinforcement Learning approach using GNNs and MAPPO for dynamic multi-agent airlift planning

Airlift Challenge: A Competition for Optimizing Cargo Delivery — *Apr. 2024*

- 2nd place in the 2024 AFRL Airlift Challenge
- Designed and implemented the core routing approach for coordinating heterogeneous aircraft across dynamic networks with bottlenecks and disruptions

Honors

Deutschlandstipendium (German National Scholarship) — Bonn, Germany
Federal Ministry of Education and Research — *Oct. 2024 & Oct. 2025*

- Merit-based federal scholarship for outstanding academic achievement

Skills

Programming Languages: Python, C++
Libraries & Frameworks: PyTorch, NumPy, Pandas, Gymnasium, NetworkX
Tools: Git, Linux, Docker, Slurm, OR-Tools
Languages: German (native), English (C1), Spanish (B1)